ANUPAMA RAY
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(a) Education & Training

Indian Institute of Technology Delhi (IITD)	Electrical Engineering	Ph.D., 2012-2016
National Institute of Technology Rourkela	Biomedical Engineering	MTech 2012
West Bengal University of Technology	Electronics Engineering	BTech 2010

(b) Research & Professional Experience

2024- present	Senior Research Scientist, IBM Research India
2017 –2024	Research Scientist, IBM Research India
2022 – present	Adjunct Professor, Indian Institute of Technology Jodhpur
2020 – 2021	Technical Assistant to Director IBM Research and CTO IBM India South-Asia

(c) Awards

- IEEE Best Women Professional Award 2021 (Early Career) from IEEE India and Hope foundation.
- Technology Rising Star 2022 Award from WOC STEM awards
- Global Volunteer Excellence Award 2020 for building AI solution to improve caregiving of children rescued from trafficking. The award money of 10K USD was donated to the NGO.
- Outstanding Technical Achievement Award at IBM in 2024, 2021, 2019, Research Division Award,IBM 2018, Eminence and Excellence Award, IBM 2018
- 4 Best paper awards (ICPR 2016, IAAI 2019, ACM MM 2019, IAAI 2020)
- Young Scientist 2018 at Global Young Scientist Summit 2018, Singapore

(d) Publications

1. Anupama Ray, et.al, Hybrid Quantum-Classical Graph Neural Networks for Tumor Classification in Digital Pathology, QCE 2024: 1611-1616
2. A. Rao, D. Madan, A. Ray, D. Vinayagamurthy, and M. S. Santhanam, Learning hard distributions with quantum-enhanced variational autoencoders (2023), arXiv:2305.01592 [quant-ph] .
3. Anupama Ray, Sai Sakunthala Guddanti, Anil Prabhakar, Improving phishing detection in Ethereum transaction network using Quantum Machine Learning. QCE 2023: 1107-1113
4. T. Khare, R. Majumdar, R. Sangle, A. Ray, P. V. Seshadri, and Y. Simmhan, Paral- lelizing quantum-classical workloads: Profiling the impact of splitting techniques, QCE 2023 https://ieeexplore.ieee.org/document/10313656
5. Divyank Tiwari, Diptesh Kanojia, Anupama Ray, Apoorva Nunna, Pushpak Bhattacharyya, Predict and Use: Harnessing Predicted Gaze to Improve Multimodal Sarcasm Detection. EMNLP 2023: 15933-15948, https://aclanthology.org/2023.emnlp-main.988/
6. A. Ray, S. Mishra, A. Nunna, and P. Bhattacharyya, Proceedings of the Thirteenth Language Resources and Evaluation Conference, LREC 2022, , 6992 (2022).
7. S. Agarwal, J. Bandlamudi, A. Mandal, A. Ray, and G. Sridhara, AI Mag. 41, 45 (2020).
8. A. Ray, P. Aggarwal, C. Hadhazi, G. Dasgupta, and A. M. Paradkar, in The Thirty-Fourth AAAI Conference on Artificial Intelligence, AAAI 2020 (AAAI Press, 2020) pp. 13196–13203.
9. S. Masihullah, R. Garg, P. Mukherjee, and A. Ray, in 25th International Conference on

Pattern Recognition, ICPR 2020 (IEEE, 2020) pp. 5812–5819.
10. A. Ray, P. Agarwal, C. K. Maurya, and G. Dasgupta, IBM J. Res. Dev. 63, 6:1 (2019).
11. A. Mandal, N. Malhotra, S. Agarwal, A. Ray, and G. Sridhara, in The Thirty-Third AAAI Conference on Artificial Intelligence, AAAI 2019 (AAAI Press, 2019) pp. 9381–9388.
12. A. Ray, M. Sharma, A. Upadhyay, M. Makwana, S. Chaudhury, A. Trivedi, A. P. Singh, and A. K. Saini, in 2019 International Conference on Document Analysis and Recognition, ICDAR 2019, Sydney, Australia, September 20-25, 2019 (IEEE, 2019) pp. 59–64.
13. A. Mandal, S. Agarwal, N. Malhotra, G. Sridhara, A. Ray, and D. Swarup, in Service-Oriented Computing - 17th International Conference, ICSOC 2019, Lecture Notes in Computer Science, Vol. 11895 (Springer, 2019) pp. 431–446.
14. A. Ray, S. Kumar, R. Reddy, P. Mukherjee, and R. Garg, in Proceedings of the 9th International on Audio/Visual Emotion Challenge and Workshop, AVEC@MM 2019 (ACM, 2019) pp. 81–88.
15. A. Gupta, A. Ray, G. Dasgupta, G. Singh, P. Aggarwal, and P. Mohapatra, in Proceedings of the 27th International Conference on Computational Linguistics, COLING 2018 (Associa- tion for Computational Linguistics, 2018) pp. 3251–3259.
16. P. Mohapatra, A. Ray, and G. Dasgupta, in Proceedings of the ACM India Joint International Conference on Data Science and Management of Data, COMAD/CODS 2018 (ACM, 2018) pp. 220–229.
17. A. Mandal, N. Malhotra, S. Agarwal, A. Ray, and G. Sridhara, in Service-Oriented Computing - 16th International Conference, ICSOC 2018, Lecture Notes in Computer Science, Vol. 11236 (Springer, 2018) pp. 332–341.
18. M. Sharma, A. Ray, S. Chaudhury, and B. Lall, in 14th IAPR International Conference on Document Analysis and Recognition, ICDAR 2017 (IEEE, 2017) pp. 466–471.
19. A. Ray, S. Rajeswar, and S. Chaudhury, in Eighth International Conference on Advances in Pattern Recognition, ICAPR 2015 (IEEE, 2015) pp. 1–6.
20. A. Ray, S. Rajeswar, and S. Chaudhury, in 13th International Conference on Document Analysis and Recognition, ICDAR 2015 (IEEE Computer Society, 2015) pp. 936–940.
21. A. Ray, S. Rajeswar, and S. Chaudhury, in 13th International Conference on Document Analysis and Recognition, ICDAR 2015 (IEEE Computer Society, 2015) pp. 1256–1260.
22. A. Ray, S. Rajeswar, and S. Chaudhury, in Proceedings of the 2014 Indian Conference on Computer Vision, Graphics and Image Processing, ICVGIP 2014 (ACM, 2014) pp. 71:1–71:8.
23. A. Ray, A. Chandawala, and S. Chaudhury, in 12th International Conference on Document Analysis and Recognition, ICDAR 2013 (IEEE Computer Society, 2013) pp. 18–22.

(e) Patents

- Reinforcement learning approach to determine the “best” setting for digital ZNE
- System and method to maximize QPU cloud utilization through opportunistic piggy-backing of queued quantum circuits
- Enhancing user experience via learning-based Quantum experimental setup selection
- System and Method to Generate and suggest Ranked Ansatz-Hardware pairs for Variational Quantum Algorithms
- System and Methods for Creating Task-specific Language Set for Multilingual Learning
- Systems and Methods for self learning annotation
- A system and Method for Automatic Adjustment of Brightness of Electronic Devices based on Visual Features using AI
- System and Method for AI based Synthesis/Transpilation of general quantum circuits.
- System And Method For Job Queue Management Using Joint Workload-Layout Quality Indicators
- A system and method to explain speech models with stylized frequency amplifications that facilitate human intuitions
- A System and Method for Consistency Verification of Product’s Information on E-commerce Portals
- A System and Method for Finding Best Accommodation using Deep Asthetic Features in Multimodal Data from Social Networks
- System and Method to infer Elements required for efficient resolution handling in Technical Support Services
- Automatic analysis and remediation of dynamic perturbations in resolver classes for retrospective learning and assignment of email tickets
- A System and Method for Consistency Verification of Product’s Information on E-commerce Portals
- A System and Method for Finding Best Accommodation using Deep Asthetic Features in Multimodal Data from Social Networks
- Automatic analysis and remediation of dynamic perturbations in resolver classes for retrospective learning and assignment of email tickets
- System and Method for automated and personalized poetry evaluation using type identification and automatic thresholding techniques
- Pro-active detection, prediction and management of technology shift in helpdesk email tickets for optimal resource utilization and productivity.

(f) Teaching Activities

- Co-Instructed a MOOC course titled “Introduction to Quantum Computing: Quantum Algorithms and Qiskit” with NPTEL with over 20K participants, running since 2021
- Teaching fullsemester courses on Quantum Information, and Quantum Machine Learning at IIT Jodhpur as an Adjunct professor
- Served as PI in IBM-IITB AI Horizon Network 2019-2021 - leading to several successful projects, publications, open-sourced datasets that were used by WatsonOneNLP and 5 masters thesis submission.
- Overall supervised 30+ Masters students for their masters thesis.